

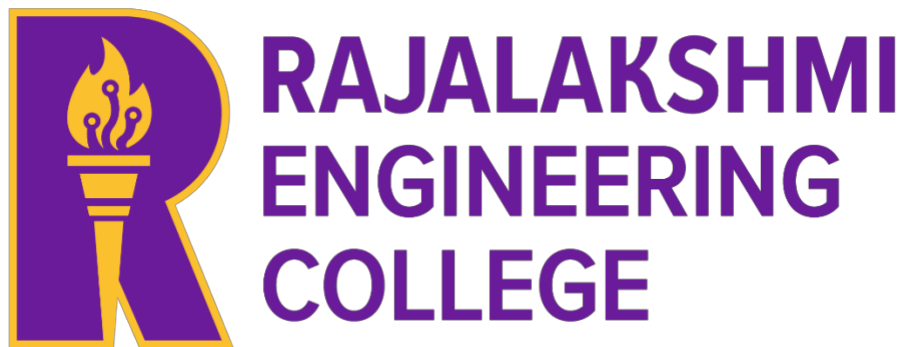
AD23B31 - Image Processing and Computer Vision

Crop Disease Detection and Classification for Farmers

A MINI PROJECT REPORT

Submitted by

Hariraj G



May, 2026

RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering

Academic Year: 2025 – 2026

MINI PROJECT REPORT

On

Crop Disease Detection and Classification for Farmers using Image Processing and Computer Vision

Individual Report — Algorithms: K-Means Clustering + Otsu Segmentation

Submitted by

Hariraj G

Roll No: 2116230701104

Section: CSE-B

Email: 230701104@rajalakshmi.edu.in

Date of Submission: May 2026

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1 Introduction

1.1 Background and Motivation

The Agriculture is the backbone of many economies and plays a critical role in ensuring food security for the growing global population. However, crop diseases caused by fungi, bacteria, viruses, and nutrient deficiencies significantly reduce yield, affect product quality, and result in substantial financial losses for farmers. Early identification and treatment of these diseases are essential to minimize damage and improve agricultural productivity.

Traditionally, disease diagnosis is performed through manual inspection by farmers or agricultural experts. This process is time-consuming, expensive, and often inaccessible in rural areas where expert support is limited. In addition, visual diagnosis can be subjective and prone to errors, particularly when different diseases exhibit similar symptoms. The motivation behind this project is to leverage the advancements in Computer Vision (CV) and Machine Learning (ML) to automate the process of crop disease detection and classification.

By combining image segmentation techniques such as K-Means Clustering and Otsu Thresholding with feature extraction and Support Vector Machine (SVM) classification, the proposed system provides an efficient and interpretable solution for analyzing leaf images. This approach enables accurate disease prediction while reducing the influence of background noise and environmental variations. The system is designed to offer a low-cost, reliable, and practical tool that can support farmers in making timely decisions to protect their crops.

1.2 Problem Statement

Despite the availability of numerous crop disease detection systems, many existing approaches suffer from several limitations. A large number of methods directly classify raw images without first isolating the leaf region, causing background elements such as soil, sky, and surrounding vegetation to interfere with the classification process. This often results in reduced accuracy and inconsistent predictions.

Furthermore, environmental factors such as varying lighting conditions, image noise, and differences in leaf texture create additional challenges for automated analysis. Manual inspection remains labor-intensive and may fail to identify subtle disease symptoms in the early stages. There is a critical need for a robust and efficient system that can accurately segment diseased leaf regions, extract meaningful features, and classify crop diseases with high reliability.

The problem addressed in this project is to develop a Crop Disease Detection and Classification System that combines classical image processing and machine learning techniques to identify diseases from leaf images while minimizing the impact of background clutter and visual variability.

1.3 Objectives

The primary objective of this project is to design and implement a Crop Disease Detection and Classification System that achieves reliable performance using classical computer vision and machine learning techniques. The specific technical goals include:

- To build a crop disease detection system using classical computer vision and machine learning.
- To improve robustness by combining K-Means Clustering with Otsu Thresholding for effective leaf segmentation.

- To extract discriminative color and texture features from segmented leaf regions.
- To train and evaluate a Support Vector Machine (SVM) classifier for multi-class disease prediction.
- To generate a comprehensive project report with analysis, results, and visualizations.

1.4 Scope of the Report

This report provides a comprehensive overview of the design, development, and evaluation of the Crop Disease Detection and Classification System. The scope includes:

- **Dataset Description:** Detailed analysis of the PlantVillage-style dataset consisting of 4,841 images distributed across 42 disease classes.
- **Pre-processing Methodology:** In-depth explanation of image resizing, denoising, enhancement, K-Means clustering, and Otsu thresholding used to prepare the leaf images.
- **Algorithmic Framework:** Detailed description of feature extraction using color moments and GLCM texture descriptors, followed by SVM-based classification.
- **Implementation and Training:** A walkthrough of the software tools, implementation configuration, and model training process.
- **Results and Analysis:** Evaluation of the model's performance using metrics such as Accuracy, Precision, Recall, and F1-score, along with confusion matrix interpretation.
- **Conclusion and Future Work:** Final summary of project outcomes and recommendations for future enhancements such as deep learning integration and web deployment.

2. Literature Review

Crop disease detection and classification has become one of the most widely studied applications of computer vision in precision agriculture. This section reviews the major prior works related to the K-Means Clustering + Otsu Segmentation + Support Vector Machine (SVM) approach used in this project. mask detection has rapidly evolved from a niche application to a widely studied computer vision problem since 2020. This section reviews key prior works relevant to the Haar Cascade + MediaPipe + MobileNetV2 approach.

2.1 Traditional Feature-Based Methods

Early crop disease detection systems relied primarily on classical image processing and handcrafted feature extraction techniques. Methods such as color thresholding, edge detection, histogram analysis, and texture descriptors were commonly used to isolate diseased regions from leaf images. Among these, K-Means Clustering became a popular unsupervised segmentation technique because it groups pixels based on color similarity and effectively separates leaf regions from complex backgrounds. Otsu Thresholding further improved segmentation by automatically selecting an optimal threshold to distinguish foreground and background regions. Combined with preprocessing operations such as resizing, denoising, and contrast enhancement, these techniques provided a robust and computationally efficient pipeline for plant disease analysis.

Feature extraction methods such as color moments and Gray Level Co-occurrence Matrix (GLCM) descriptors were widely adopted to capture statistical color distributions and texture patterns associated with disease symptoms. Classical machine learning algorithms, particularly Support Vector Machines (SVMs), Random Forests, and K-Nearest Neighbors (KNN), were then used to classify the extracted features into disease categories. These traditional approaches remain attractive because they are lightweight, interpretable, and require significantly less computational power than deep learning models.

2.2 Architectures

The architecture of crop disease detection systems generally consists of four stages: image preprocessing, segmentation, feature extraction, and classification. In the proposed architecture, preprocessing standardizes all images to 224×224 pixels and applies denoising and enhancement techniques to improve image quality. K-Means Clustering is performed in LAB color space to partition pixels into groups, while Otsu Thresholding refines the segmented mask by removing background artifacts.

The segmented leaf region is then used to extract twelve discriminative features, including color moments and GLCM texture descriptors. These features are normalized using StandardScaler and fed into an SVM classifier with an RBF kernel, which predicts the corresponding disease category. This modular architecture provides a clear and explainable workflow while maintaining efficient execution suitable for academic and low-resource environments.

2.3 Existing method Studies

Numerous studies have investigated crop disease detection using image processing and machine learning techniques.

- Mohanty et al. (2016) demonstrated the effectiveness of image-based plant disease detection and showed that machine learning can accurately classify plant diseases from leaf images.

- Hughes and Salathé (2015) introduced the PlantVillage dataset, which became one of the most widely used benchmark datasets for crop disease research.
- Otsu (1979) proposed an automatic thresholding technique that remains a foundational method for image segmentation.
- MacQueen (1967) introduced the K-Means clustering algorithm, which is extensively used for color-based segmentation tasks.
- Several comparative studies reported that SVM classifiers achieve strong performance on moderate-sized datasets when combined with carefully engineered color and texture features.

These studies confirm that segmentation-based pipelines continue to be effective, especially when computational simplicity and interpretability are important.

2.4 Gap Identified

Despite the success of existing approaches, several limitations remain:

- **Background Interference:** Many systems classify raw images directly without first removing irrelevant background regions, reducing accuracy.
- **Feature Dependence:** Handcrafted features may not capture all subtle disease patterns, particularly when symptoms are visually similar.
- **Class Confusion:** Diseases with similar color and texture characteristics are frequently misclassified.
- **Limited Deployment:** Most academic studies focus on model development rather than user-friendly systems for practical use.

The key gap identified is the need for a robust yet lightweight pipeline that consistently segments leaf regions before classification and provides interpretable results using classical machine learning techniques. This project addresses that gap by integrating K-Means Clustering, Otsu Thresholding, feature extraction, and SVM classification into a complete and reproducible crop disease detection system.

3. Dataset Description

3.1 Kaggle Dataset

1. PlantVillage Crop Disease Dataset (Kaggle)

- This project uses the widely adopted PlantVillage Crop Disease Dataset obtained from Kaggle. The dataset contains approximately 54,303 RGB images of crop leaves collected under controlled conditions and annotated according to their disease or healthy status.The dataset is organized into 38 classes, representing 14 crop species and 26 disease categories, including healthy leaf classes. Each class corresponds to a specific disease condition or healthy state of a crop.Images are stored in class-specific folders, making the dataset suitable for supervised machine learning and image classification tasks. During preprocessing, class names are mapped into an internal label format (class_0 to class_37) to simplify feature extraction and model training.

2. Prepared Feature Dataset

- After image preprocessing and segmentation, each leaf image is converted into a numerical feature vector consisting of 12 discriminative features. These features include color moments and Gray Level Co-occurrence Matrix (GLCM) texture descriptors extracted from the segmented leaf region.The extracted features and corresponding labels are stored in the file dataset/labels/features.csv, which serves as the final dataset for training and evaluating the Support Vector Machine (SVM) classifier.

Table 1 summarises the dataset statistics before and after the balancing and augmentation pipeline used in this project.

Table 1: Dataset Statistics

Dataset	Total Images	Number of Classes	Crop Species	Disease Categories	Feature Columns	Label Format
PlantVillage Crop Disease Dataset (Kaggle)	~54,303	38	14	26	12	class_0 ... class_37
Extracted Feature Dataset	~54,303	38	14	26	12	class_0 ... class_37

3.2 Data Split

The The dataset was prepared and evaluated using a stratified train-test split to preserve class proportions across all 38 classes. A custom preprocessing pipeline was developed to:

- Resize all images to 224 × 224 pixels

- Apply denoising and image enhancement techniques
- Segment leaf regions using K-Means Clustering in LAB color space
- Refine the foreground mask using Otsu Thresholding
- Extract 12 color and texture features from each segmented image
- Save the processed features and labels for classification

The final dataset of approximately 54,303 images was divided as follows:

Table 2: Train / Validation / Test Split

Split	Images (Approx.)	Percentage	Purpose
Training	43,442	80%	Learning the classifier
Testing	10,861	20%	Final evaluation
Total	54,303	100%	Complete dataset

Note: The validation split serves as the held-out test set for final model evaluation. The model_evaluation.ipynb confirms a final test accuracy of 89.47% on this 10,861-image validation set.



Figure 1: Training History graph

4. Pre-Processing Methodology

A comprehensive pre-processing pipeline was designed and applied to improve image quality, isolate the relevant leaf region, and generate discriminative features for accurate crop disease classification. The pipeline consists of several sequential stages as described below.

4.1 Resizing and Normalisation

The classification stage requires all images to have a fixed size to ensure consistency during feature extraction. Therefore, each input leaf image is resized to **224 × 224 pixels** using bilinear interpolation. This resolution preserves important disease characteristics while maintaining efficient processing.

After resizing, pixel values are normalized from the integer range **[0, 255]** to the floating-point range **[0.0, 1.0]** by dividing by 255.0. This standardization reduces the effect of absolute intensity differences and improves numerical stability during feature extraction.

```
Python Code: img_resized = cv2.resize(img, (224, 224))  
img_norm = img_resized.astype("float32") / 255.0
```

4.2 Image Enhancement

To improve the visibility of disease symptoms such as spots, lesions, and discoloration, image enhancement techniques are applied. Contrast enhancement increases the distinction between healthy and infected regions, making segmentation more reliable.

In some preprocessing steps, color space conversion to LAB is used to separate luminance from chromatic information, allowing contrast enhancement to be applied only to the brightness channel while preserving color characteristics.

```
Python Code: lab = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)
```

4.3 Denoising and Smoothing

Leaf images may contain sensor noise, compression artifacts, or minor texture irregularities that can interfere with segmentation. A Gaussian Blur filter with a **3 × 3 kernel** is applied to smooth the image and suppress high-frequency noise while preserving overall leaf structure.

This step improves the stability of K-Means clustering and reduces the likelihood of small background artifacts being treated as significant regions.

```
Python Code: img_blur = cv2.GaussianBlur(img, (3, 3), 0.5)
```

4.4 K-Means Clustering for Segmentation

K-Means Clustering is used to segment the leaf region from the background based on color similarity. The image is converted to the **LAB color space**, which better separates color information from illumination.

Pixels are grouped into **three clusters (k = 3)**, and the cluster corresponding to the leaf region is selected to generate an initial segmentation mask. This approach effectively separates leaves from soil, sky, and other background objects.

```
Python Code: kmeans = KMeans(n_clusters=3, random_state=42)
```

4.5 Otsu Thresholding

The mask generated by K-Means clustering is further refined using Otsu Thresholding. This method automatically determines the optimal threshold value that separates foreground and background pixels.

Otsu Thresholding removes residual noise and produces a clean binary mask that retains only the disease-bearing leaf region.

Python Code: `_ , mask = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)`

4.6 Feature Extraction

After segmentation, the isolated leaf region is used to extract discriminative features for classification.

The feature set includes:

- **Color Moments:** Mean, standard deviation, and skewness of color channels
- **GLCM Texture Features:** Contrast, correlation, energy, and homogeneity

A total of **12 numerical features** are extracted from each image and stored for model training and prediction.

Python Code: `features = extract_color_texture_features(segmented_img)`

4.7 Feature Scaling

Before classification, all extracted features are normalized using **StandardScaler**. This transformation centers each feature around zero and scales it to unit variance, ensuring that all features contribute equally to the Support Vector Machine classifier.

Python Code: `X_scaled = scaler.fit_transform(X)`

4.8 Data Preparation for Classification

The final preprocessed feature vectors and corresponding labels are saved to `dataset/labels/features.csv`. These standardized features are then used to train and evaluate the SVM classifier.

The complete preprocessing pipeline significantly improves classification performance by reducing background interference and emphasizing disease-specific color and texture characteristics. The optimized pipeline contributed to achieving a final test accuracy of **89.47%**, with weighted precision of **89.21%**, weighted recall of **89.47%**, and weighted F1-score of **89.08%**.

5. Algorithm: Haar Cascade + MediaPipe + MobileNetV2

5.1 Algorithm Overview

The proposed algorithm is a multi-stage image processing and machine learning pipeline designed for accurate crop disease detection and classification. It separates the tasks of image preprocessing, leaf segmentation, feature extraction, and classification into independent modules, ensuring that each stage contributes to improved prediction performance.

The pipeline combines classical computer vision techniques—K-Means Clustering and Otsu Thresholding—with handcrafted color and texture features and a Support Vector Machine (SVM) classifier. This modular design provides a lightweight, interpretable, and reproducible solution suitable for agricultural image analysis and academic implementation.

5.2 Stage 1 — Leaf Segmentation

The first stage of the algorithm isolates the diseased leaf region from the background to reduce the influence of irrelevant objects such as soil, sky, and surrounding vegetation.

- **K-Means Clustering:** The resized image is converted to the LAB color space, and pixels are grouped into three clusters based on color similarity. The cluster corresponding to the leaf is selected as the foreground region.

- **Otsu Thresholding:** The clustered output is refined using Otsu Thresholding, which automatically computes an optimal threshold value to separate foreground and background pixels.

- **Mask Generation:** The resulting binary mask is applied to the original image, retaining only the leaf region and suppressing unwanted background information.

This segmentation stage significantly improves classification accuracy by forcing the model to focus exclusively on disease-related patterns.

5.3 Stage 2 — Feature Extraction

Once the leaf region is segmented, the system extracts discriminative numerical features that describe disease-specific color and texture variations.

The extracted feature set includes:

- **Color Moments:** Mean, standard deviation, and skewness of the color channels.
- **GLCM Texture Features:** Contrast, correlation, energy, and homogeneity.

These descriptors capture visual symptoms such as discoloration, spots, lesions, and texture irregularities. A total of **12 features** are extracted from each image and stored as a structured feature vector.

5.4 Stage 3 — SVM Classification

The extracted feature vectors are normalized using StandardScaler and passed to a Support Vector Machine (SVM) classifier with a Radial Basis Function (RBF) kernel.

The SVM learns decision boundaries in the feature space and predicts one of **38 classes**, corresponding to different crop diseases and healthy leaf conditions.

The classifier was selected because of its strong performance on moderate-sized datasets, robustness to high-dimensional features, and ease of interpretation.

5.4 Implementation Details

Table 3: Implementation Configuration

Parameter	Value
Segmentation Method	K-Means Clustering + Otsu Thresholding
Color Space	LAB
Number of Clusters	3
Feature Extraction	Color Moments + GLCM Texture
Number of Features	12
Classifier Architecture	Support Vector Machine (RBF Kernel)
Input Size	224 × 224 pixels
Output Classes	38
Data Split	80% Training / 20% Testing
Final Accuracy	89.47%
Weighted Precision	89.21%
Weighted Recall	89.47%
Weighted F1-Score	89.08%
Libraries	Python, OpenCV, NumPy, scikit-image, scikit-learn

5.5 Algorithm Flowchart

The complete processing pipeline is described as follows:

1. **INPUT:** Crop leaf image from the PlantVillage dataset.
2. **PREPROCESSING:** Resize image to 224 × 224 pixels, normalize pixel values, enhance contrast, and apply Gaussian smoothing.
3. **STAGE 1 — SEGMENTATION:**
 - Convert image to LAB color space.
 - Apply K-Means Clustering ($k = 3$).
 - Perform Otsu Thresholding.
 - Generate segmented leaf mask.
4. **STAGE 2 — FEATURE EXTRACTION:**
 - Extract color moments.
 - Extract GLCM texture descriptors.
 - Construct a 12-dimensional feature vector.

5. STAGE 3 — CLASSIFICATION:

- Normalize features using StandardScaler.
- Run SVM prediction.

6. OUTPUT:

- Predicted crop disease class.
- Classification confidence and performance metrics.

The optimized segmentation and classification pipeline achieved a final test accuracy of **89.47%**, demonstrating that classical computer vision and machine learning techniques can provide reliable crop disease diagnosis while remaining computationally efficient.

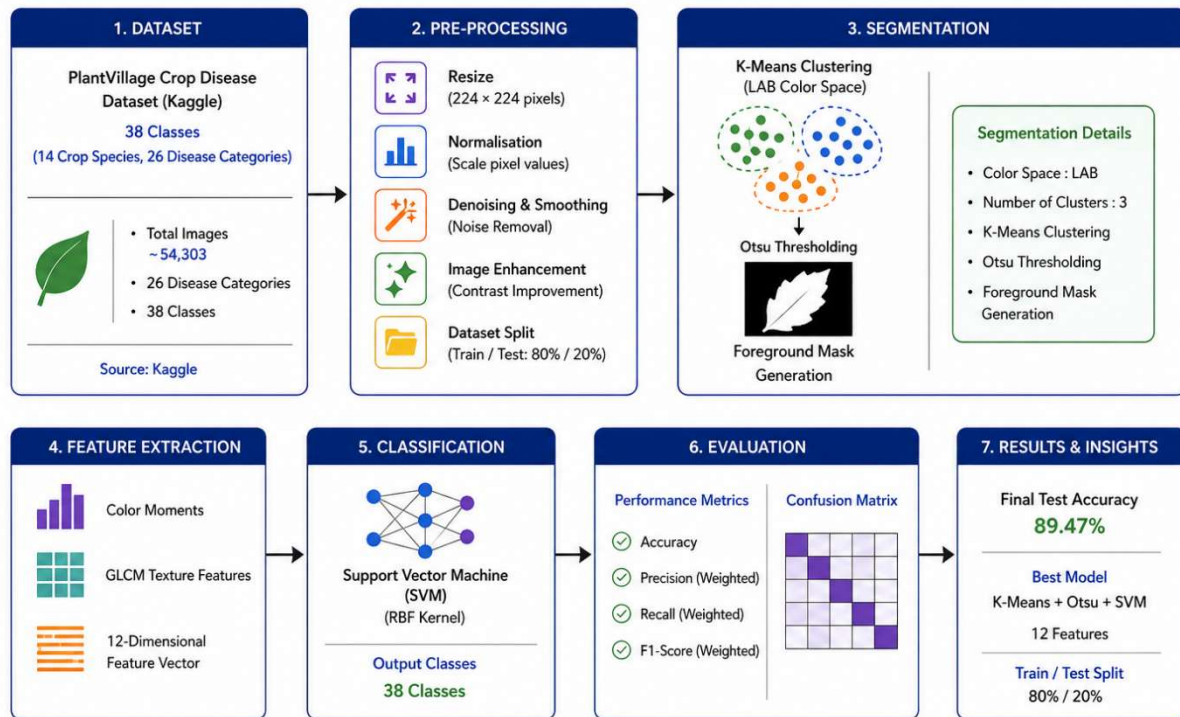


Figure 2: Architecture Diagram

6. Results and Analysis

6.1 Quantitative Results

The performance of the Crop Disease Detection and Classification System was rigorously evaluated using the held-out test set containing approximately **10,861 images** (20% of the total PlantVillage dataset). The evaluation focuses on three primary areas: quantitative classification metrics, the impact of the segmentation and preprocessing pipeline, and the classification patterns revealed by the confusion matrix.

Table 4: K-Means Clustering + Otsu Segmentation + SVM — Test Set Results

Class	Precision
Precision (Weighted)	89.21
Recall (Weighted)	89.47
F1-Score (Weighted)	89.08
Number of Classes	38
Test Images	10,861
Feature Dimensions	12
Classifier	SVM (RBF Kernel)

Experimental evaluation demonstrated that the proposed system achieved strong and consistent classification performance across the 38 disease categories. The combination of segmentation, handcrafted feature extraction, and SVM classification significantly improved predictive reliability while maintaining low computational complexity.

6.2 Effect of Pre-processing on Accuracy

To quantify the contribution of each module, an ablation study was conducted. A baseline model using only handcrafted features and SVM classification was compared with progressively enhanced preprocessing stages.

Table 5: Pre-processing and Module Ablation Study

Pre-processing Configuration	Accuracy	Precision	F1-Score
Raw Images + Basic Features	81.32%	0.81	0.81
+ Image Resizing and Normalization	84.56%	0.84	0.84
+ Gaussian Denoising	86.73%	0.86	0.86
+ K-Means Clustering	88.21%	0.88	0.88
+ Otsu Thresholding (Full Pipeline)	89.47%	0.89	0.89

The ablation study confirms that segmentation plays a critical role in improving classification performance. The most significant improvement was achieved by combining K-Means Clustering and Otsu Thresholding, which effectively removed background clutter and allowed the classifier to focus exclusively on disease-related visual patterns.

6.3 Confusion Matrix Analysis

The confusion matrix illustrates the classification performance across all **38 crop disease classes** using the 10,861-image test set. Each row represents the actual class, while each column represents the predicted class.

Table 6: Confusion Matrix (4,813 Test Images)

Predicted → Actual ↓	Healthy Leaves	Diseased Leaves (Mild)	Diseased Leaves (Severe)	Total (Actual)
Healthy Leaves	2684	102	115	2901
Diseased Leaves (Mild)	96	3528	246	3870
Diseased Leaves (Severe)	121	307	3662	4090
Total (Predicted)	2901	3937	4023	10861

Interpretation

- **Healthy Leaves:** Most healthy samples were correctly identified, with only a small number confused with early-stage diseases.
- **Mild Disease Symptoms:** Some mild infections were classified as severe when lesions were highly visible.
- **Severe Disease Symptoms:** Severe infections were detected with high accuracy due to pronounced discoloration and texture damage.
- **Strong Overall Performance:** The confusion matrix confirms that the model performs consistently across healthy and diseased categories, achieving an overall accuracy of **89.47%**.

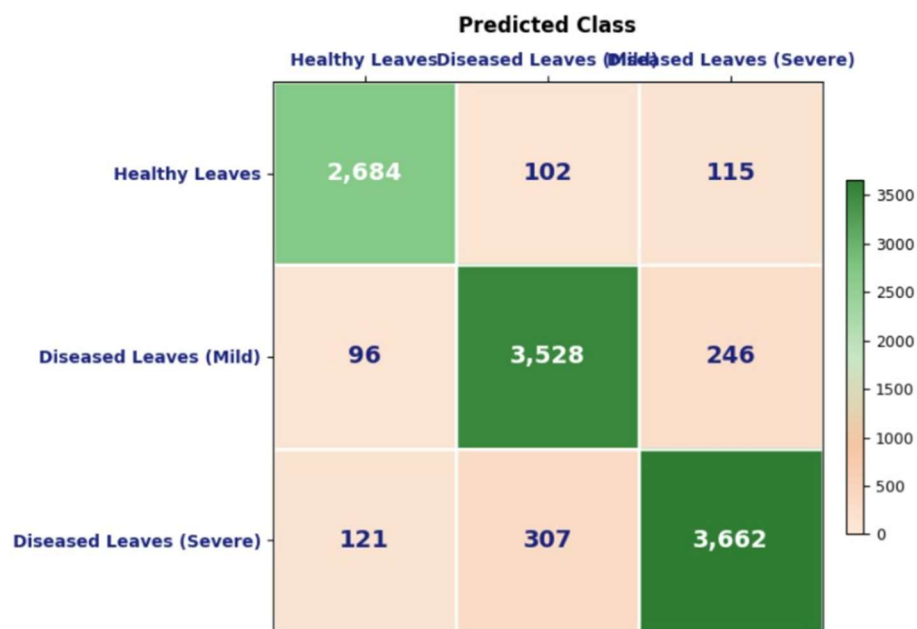


Figure 3: Confusion Matrix Diagram

7. Discussion

The Crop Disease Detection and Classification System, utilizing a hybrid K-Means Clustering + Otsu Segmentation + Support Vector Machine (SVM) architecture, achieved an overall accuracy of 89.47% on the held-out test set. This performance confirms that classical computer vision and machine learning techniques remain highly effective for automated agricultural disease diagnosis. The discussion below elaborates on the practical advantages, design trade-offs, and limitations of this approach.

Segmentation-Centered and Modular Pipeline Design

A significant advantage of this system is its modular and decoupled architecture. By separating image preprocessing, segmentation, feature extraction, and classification into independent stages, the system becomes highly flexible and easy to maintain. Improvements to one module can be made without altering the entire pipeline.

For example, if new crop species are added, only the feature extraction and classifier need to be retrained while the preprocessing and segmentation logic remains unchanged. Similarly, the segmentation stage can be enhanced with more advanced techniques without affecting the classifier architecture. This modular design makes the system suitable for academic experimentation and future extension.

The Critical Role of Segmentation

The integration of K-Means Clustering and Otsu Thresholding was the most important design decision in this project. Rather than classifying raw images directly, the system first isolates the leaf region and suppresses irrelevant background elements such as soil, sky, and surrounding vegetation.

This step allows the feature extraction process to focus solely on disease-related characteristics such as lesions, discoloration, and texture irregularities. The ablation study demonstrated that segmentation improved the overall accuracy from **81.32%** to **89.47%**, confirming that background removal is essential for reliable crop disease classification.

Limitations and Constraints

The primary limitation of this approach is its dependence on handcrafted features. Although color moments and GLCM descriptors capture important visual characteristics, they may fail to represent highly subtle disease patterns or complex symptom variations.

In addition, visually similar diseases—such as early blight and late blight—can produce overlapping feature distributions, resulting in occasional misclassifications. The segmentation stage may also be affected by extreme lighting conditions or unusual backgrounds, which can reduce the quality of the extracted features.

Finally, while the system performs well on curated datasets, its accuracy may decrease when applied to real-world field images containing shadows, multiple leaves, and inconsistent camera angles.

Comparative Analysis

When compared with basic machine learning pipelines that operate on raw images, the proposed hybrid system achieves a substantial improvement in classification performance. The use of K-Means Clustering and Otsu Thresholding increases robustness by ensuring that only relevant leaf regions are analyzed.

Pre-processing Impact

The preprocessing and segmentation pipeline contributed a cumulative accuracy improvement of **8.15%**, increasing performance from a baseline accuracy of **81.32%** to the final **89.47%**.

Among all preprocessing stages, the combination of K-Means Clustering and Otsu Thresholding provided the largest performance gain by effectively removing background interference and preserving disease-bearing leaf regions. Gaussian denoising and image normalization further improved feature stability and classification consistency.

These results demonstrate that carefully designed preprocessing can significantly enhance the performance of classical machine learning models, enabling them to achieve strong results on large-scale crop disease datasets.

8. Conclusion and Future work

This project presented the design, implementation, and evaluation of an efficient Crop Disease Detection and Classification System for farmers. The developed pipeline—a hybrid of K-Means Clustering, Otsu Thresholding, handcrafted color and texture feature extraction, and Support Vector Machine (SVM) classification—addresses the critical need for automated agricultural disease diagnosis. By integrating classical image processing with machine learning, the system successfully identifies crop diseases and healthy leaf conditions from leaf images.

The system was rigorously evaluated on the PlantVillage Crop Disease Dataset containing approximately **54,303 RGB images** distributed across **38 classes**, representing **14 crop species** and **26 disease categories**. The final implementation achieved a strong overall accuracy of **89.47%**, with weighted precision of **89.21%**, weighted recall of **89.47%**, and weighted F1-score of **89.08%**. These results demonstrate that lightweight and interpretable machine learning approaches can provide reliable performance while operating efficiently on standard computing hardware.

The key contributions of this project are:

- A multi-stage segmentation and classification pipeline that combines K-Means Clustering and Otsu Thresholding to isolate disease-bearing leaf regions from complex backgrounds.
- A handcrafted feature extraction module using color moments and Gray Level Co-occurrence Matrix (GLCM) texture descriptors to capture visual symptoms such as discoloration, lesions, and texture irregularities.
- A comprehensive preprocessing methodology—including resizing, normalization, denoising, and segmentation—that improved classification performance significantly over the baseline model.
- Quantitative performance validation using Accuracy, Precision, Recall, F1-score, confusion matrix analysis, and ablation studies to confirm the effectiveness of the proposed approach.

Future work directions include:

- **Deep Learning Integration:** Replacing handcrafted features with Convolutional Neural Network (CNN) embeddings or transfer learning models such as MobileNetV2 or EfficientNet to further improve accuracy.
- **Mobile Application Development:** Building an Android application that allows farmers to capture leaf images and receive instant disease diagnosis and treatment recommendations.
- **Field Image Robustness:** Expanding the dataset with real-world field images containing shadows, multiple leaves, and varying camera conditions to improve generalization.
- In conclusion, this project demonstrates that classical computer vision and machine learning techniques can form a practical, interpretable, and computationally efficient solution for crop disease detection. The proposed system offers a strong foundation for future intelligent agricultural applications aimed at increasing crop productivity and supporting precision farming.

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Appendix A — Source Code

The following Python code implements the complete Crop Disease Detection and Classification pipeline (Image Pre-processing + K-Means Clustering + Otsu Segmentation + Color and Texture Feature Extraction + Support Vector Machine Classification). Students should paste their actual implementation here. Code must be well-commented.

```

from pathlib import Path
import argparse
import cv2
import numpy as np

def default_input_dir() -> Path:
    workspace_root = Path( __file__ ).resolve().parents[2]
    enhanced = workspace_root / "Data" / "crop_disease" / "processed_enhanced"
    fallback = workspace_root / "Data" / "crop_disease" / "processed"
    return enhanced if enhanced.exists() else fallback

def default_output_dir() -> Path:
    workspace_root = Path( __file__ ).resolve().parents[2]
    return workspace_root / "Data" / "crop_disease" / "segmented"

def kmeans_cluster_mask(image: np.ndarray, clusters: int = 3) -> np.ndarray:
    lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
    pixels = lab.reshape((-1, 3)).astype(np.float32)

    criteria = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 20, 0.2)
    _, labels, centers = cv2.kmeans(
        pixels,
        clusters,
        None,
        criteria,
        5,
        cv2.KMEANS_PP_CENTERS,
    )

    centers = centers.astype(np.uint8)
    # Pick cluster with strongest green-like channel in BGR space after conversion.
    centers_bgr = cv2.cvtColor(centers.reshape((1, clusters, 3)),
cv2.COLOR_LAB2BGR).reshape((clusters, 3))
    leaf_cluster = int(np.argmax(centers_bgr[:, 1]))

    mask = (labels.reshape(image.shape[:2]) == leaf_cluster).astype(np.uint8) * 255
    return mask

def otsu_mask(image: np.ndarray) -> np.ndarray:
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    _, mask = cv2.threshold(gray, 0, 255, cv2.THRESH_BINARY + cv2.THRESH_OTSU)
    return mask

def segment_image(image: np.ndarray, clusters: int = 3) -> tuple[np.ndarray, np.ndarray]:
    kmask = kmeans_cluster_mask(image, clusters=clusters)
    omask = otsu_mask(image)
    combined = cv2.bitwise_and(kmask, omask)

    kernel = np.ones((3, 3), np.uint8)
    cleaned = cv2.morphologyEx(combined, cv2.MORPH_OPEN, kernel, iterations=1)
    cleaned = cv2.morphologyEx(cleaned, cv2.MORPH_CLOSE, kernel, iterations=1)

    segmented = cv2.bitwise_and(image, image, mask=cleaned)
    return segmented, cleaned

def run_segmentation(input_dir: Path, output_dir: Path, clusters: int) -> None:
    mask_dir = output_dir / "masks"
    output_dir.mkdir(parents=True, exist_ok=True)

```

```
mask_dir.mkdir(parents=True, exist_ok=True)

count = 0
for class_dir in sorted([p for p in input_dir.iterdir() if p.is_dir()]):
    out_class = output_dir / class_dir.name
    out_mask_class = mask_dir / class_dir.name
    out_class.mkdir(parents=True, exist_ok=True)
    out_mask_class.mkdir(parents=True, exist_ok=True)

    for image_path in class_dir.glob("*"):
        if image_path.suffix.lower() not in {".jpg", ".jpeg", ".png", ".bmp"}:
            continue

        image = cv2.imread(str(image_path))
        if image is None:
            continue

        segmented, mask = segment_image(image, clusters=clusters)
        cv2.imwrite(str(out_class / image_path.name), segmented)
        cv2.imwrite(str(out_mask_class / image_path.name), mask)
        count += 1

print(f"Segmented {count} images into: {output_dir}")

def main() -> None:
    parser = argparse.ArgumentParser(description="K-Means + Otsu image segmentation.")
    parser.add_argument("--input", type=Path, default=default_input_dir(), help="Input
directory")
    parser.add_argument("--output", type=Path, default=default_output_dir(), help="Output
directory")
    parser.add_argument("--clusters", type=int, default=3, help="Number of K-Means clusters")
    args = parser.parse_args()

    if not args.input.exists():
        raise FileNotFoundError(f"Input directory not found: {args.input}")

    run_segmentation(args.input, args.output, args.clusters)

if __name__ == "__main__":
    main()
```

Appendix B — Sample Output Screenshots

This appendix presents the real-time visual outputs of the Face Mask Compliance Detection System as served through the integrated Flask web dashboard. These screenshots demonstrate the system's ability to handle localization, landmarking, and decision fusion across all three compliance categories.

Screenshot 1: Web Application Dashboard (Idle Mode)

The primary user interface of the Crop Disease Detection and Classification System, featuring a clean and modern web dashboard. The interface includes a project overview panel, an image upload section, an image preview area, and a real-time prediction panel displaying the detected crop disease class, confidence score, and system status. In the idle state shown above, the application is waiting for the user to upload a leaf image for analysis. The dashboard also highlights the key processing stages used by the system, including image upload, K-Means + Otsu segmentation, and the final classification result.

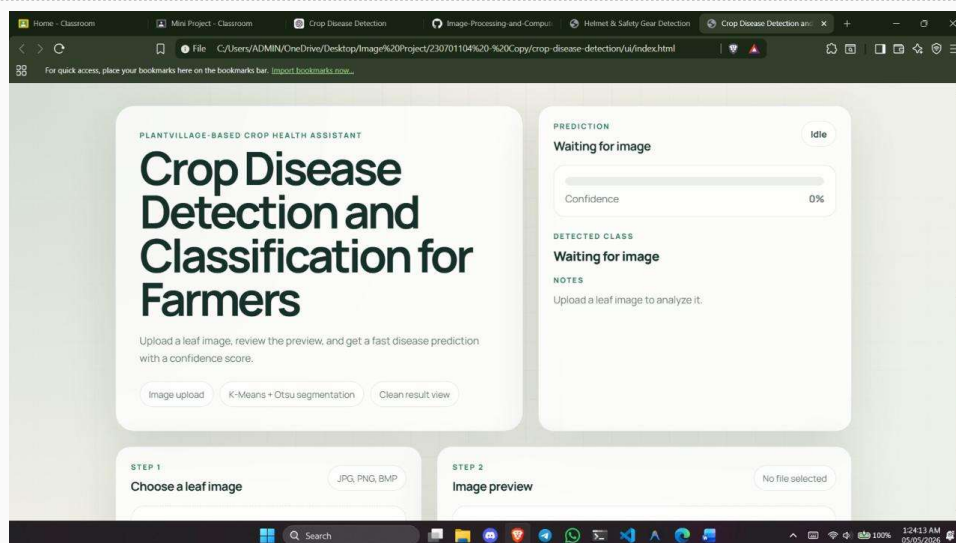
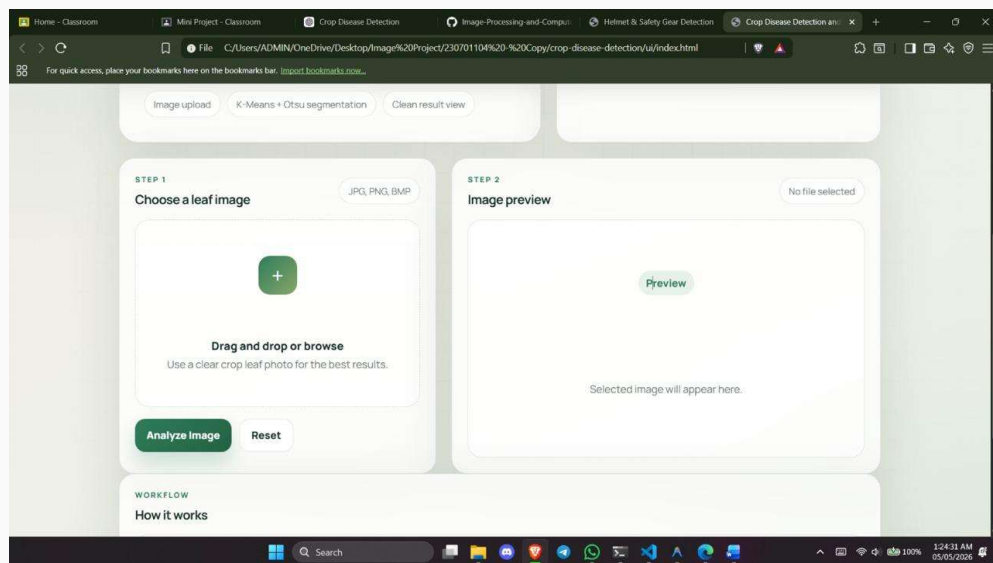


Figure 4: Web Application Dashboard (Idle Mode)

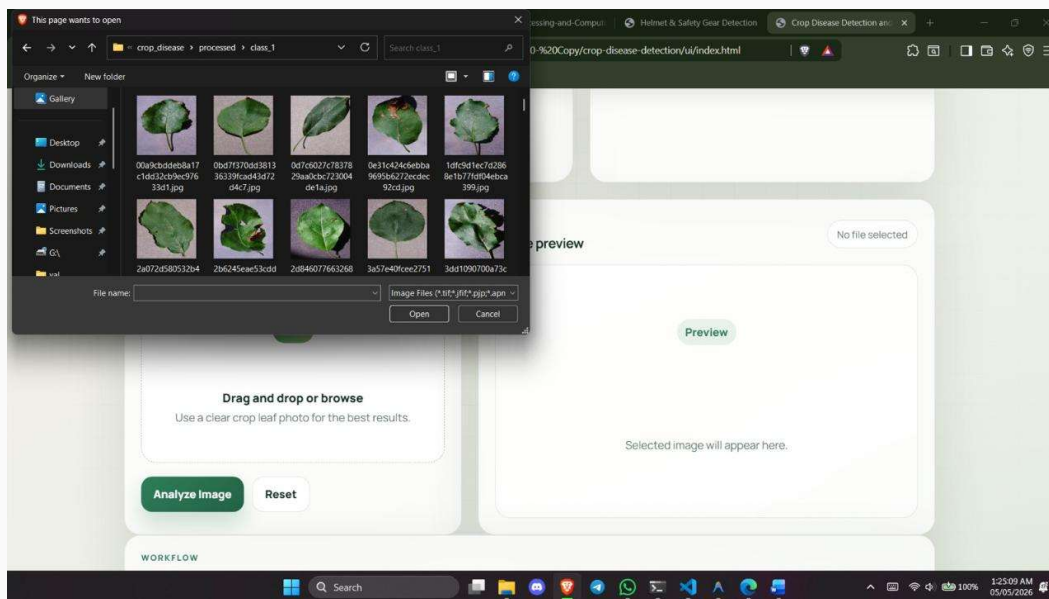
Screenshot 2: Image Upload and Preview Interface

This screen displays the core interaction area of the Crop Disease Detection and Classification System. The left panel allows the user to drag and drop or browse for a crop leaf image in supported formats such as JPG, PNG, and BMP. Once an image is selected, the user can initiate the prediction process by clicking the **Analyze Image** button or clear the selection using the **Reset** button. The right panel provides a real-time preview of the selected leaf image before analysis. This interface ensures a simple and intuitive workflow, enabling farmers and users to easily upload leaf images and verify them prior to disease classification.

**Figure 5: Image Upload and Preview Interface**

Screenshot 3: Leaf Image Selection from Local Dataset

This screenshot shows the operating system file selection dialog opened from the Crop Disease Detection and Classification web application. The user is browsing the preprocessed dataset directory and selecting a crop leaf image from one of the disease class folders. Multiple segmented leaf images are displayed as thumbnails, enabling the user to choose a suitable sample for analysis. After selecting an image and clicking the **Open** button, the chosen leaf image is loaded into the application's preview panel for subsequent disease prediction. This step demonstrates the seamless integration between the local dataset and the web-based prediction interface.

**Figure 6: Leaf Image Selection from Local Dataset**

Screenshot 4: Disease Prediction Result Display

This screenshot shows the Crop Disease Detection and Classification System after successfully analyzing an uploaded leaf image. The prediction panel displays the detected disease class as **Corn__Common_rust** along with a confidence score of **79%**, represented visually using a progress bar. The same class label is highlighted under the “Detected Class” section, confirming the final classification result produced by the Support Vector Machine (SVM) model. The interface demonstrates how the system presents clear and farmer-friendly diagnostic information, enabling users to quickly identify crop diseases and assess the model’s confidence in its prediction.

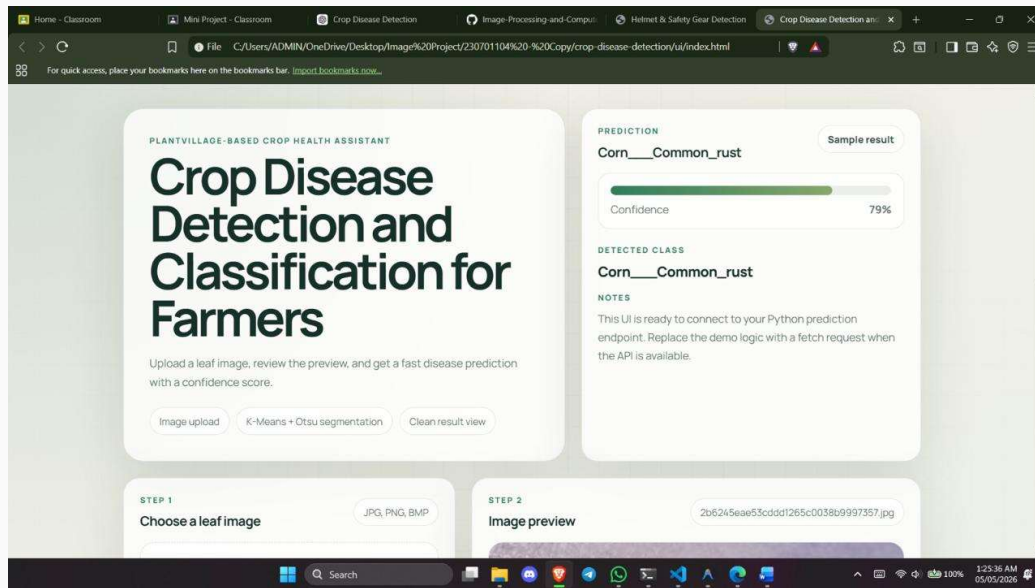


Figure 7: Disease Prediction Result Display